

EPQ Appendix Pack

This pack brings the six main appendix items into one order. It points to the maintained evidence files and includes the key visuals and summary tables.

1. Detailed Time Table

Stage	Planned period	What and why	Evidence or status
1	13-17 Jun	Define a narrow question before choosing sources or methods	Proposal history and PL-05/PL-06
2	17-23 Jun	Build and evaluate the research base	Sources, literature notes, search log and evaluation files
3	17-30 Jun	Collect checked market data and build the first common dataset	Raw/processed data, metadata and quality report
4	22 Jun-13 Jul	Implement benchmarks and the three supervised comparators	Model package, configurations and outputs
5	1-20 Jul	Compare, audit, correct and rerun the experiment	Dated predictions, automated tests and robustness outputs
6	4-26 Jul	Draft the report, revise the question and strengthen the method and evaluation	English/Chinese reports and PL-11.07
7	6 Aug	Complete Appendix checklist and document-layout QA	This pack, PL-11.08 and rebuilt bilingual DOCX files
8	26 Jul-31 Aug	Rehearse, present and complete authenticated records	Preparation exists; real delivery and signatures remain outstanding
9	20 Aug	Refresh completed candles and rerun the frozen-cutoff experiment	1,271 completed candles through 19 Aug; 950 training and 276 test rows; all overall ranks retained

The full phase-by-phase timetable, including limitations on historical date authentication, is in [appendix/timetable.md](#) .

2. Gantt Chart

EPQ Project Gantt Chart

Recorded plan and revision periods - updated 20 August 2026



The chart visualises recorded plans and revisions. It is not independent proof that every task was completed on its target date.

3. Source Evaluation

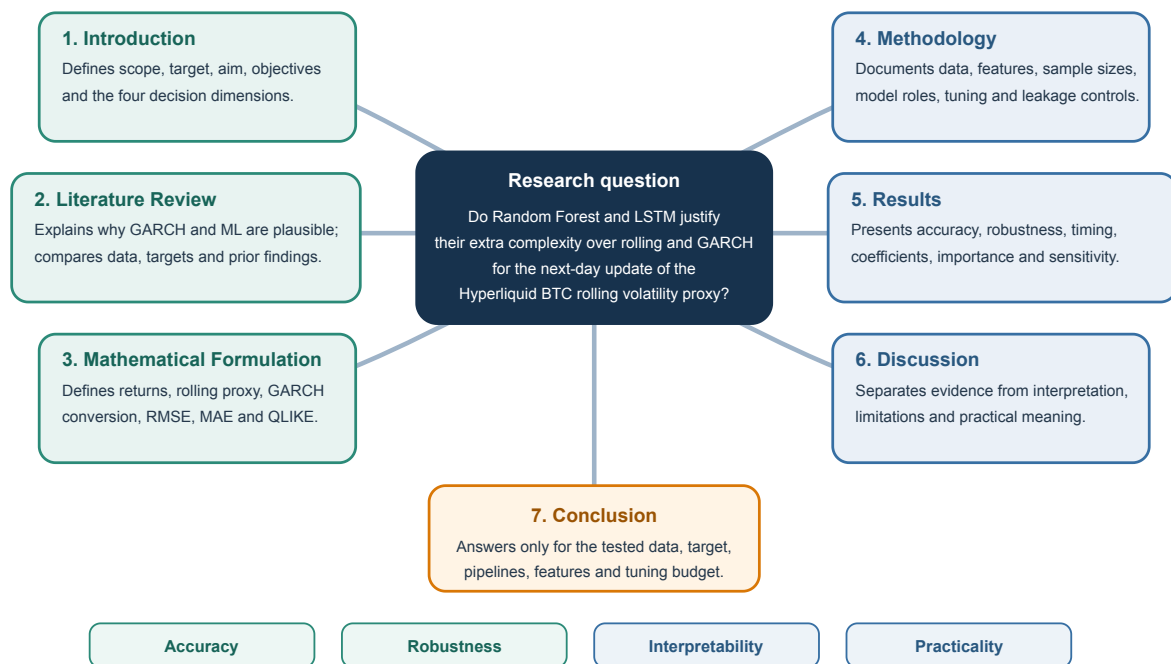
Source	Use in this EPQ	Important limitation or design difference
Hansen and Lunde (2005)	Treats GARCH(1,1) as a serious benchmark	Exchange-rate/equity evidence, not cryptocurrency or this proxy
Dudek et al. (2024)	Shows rankings vary by asset, horizon and metric	Uses richer intraday realised variance
Huang, Sangiorgi and Urquhart (2024)	Prevents a universal anti-neural-network conclusion	High-frequency inputs and more advanced architectures
Shen, Wan and Leatham (2021)	Motivates high-volatility regime checks	Different target, recurrent model and validation design
Patton (2011)	Motivates supplementary variance-scale QLIKE	Assumptions do not make this overlapping proxy latent volatility
Breiman (2001); Hochreiter and Schmidhuber (1997)	Justify RF and LSTM model inclusion	Foundational papers do not validate this implementation or dataset

The complete reconciled evaluation is in `research/source-evaluation-summary.md` ; the original workbook remains at `research/Source_Evaluation_Tianlin_He.xlsx` .

4. Mind Map for Report Structure

Report Structure Mind Map

How each section contributes to the final judgment

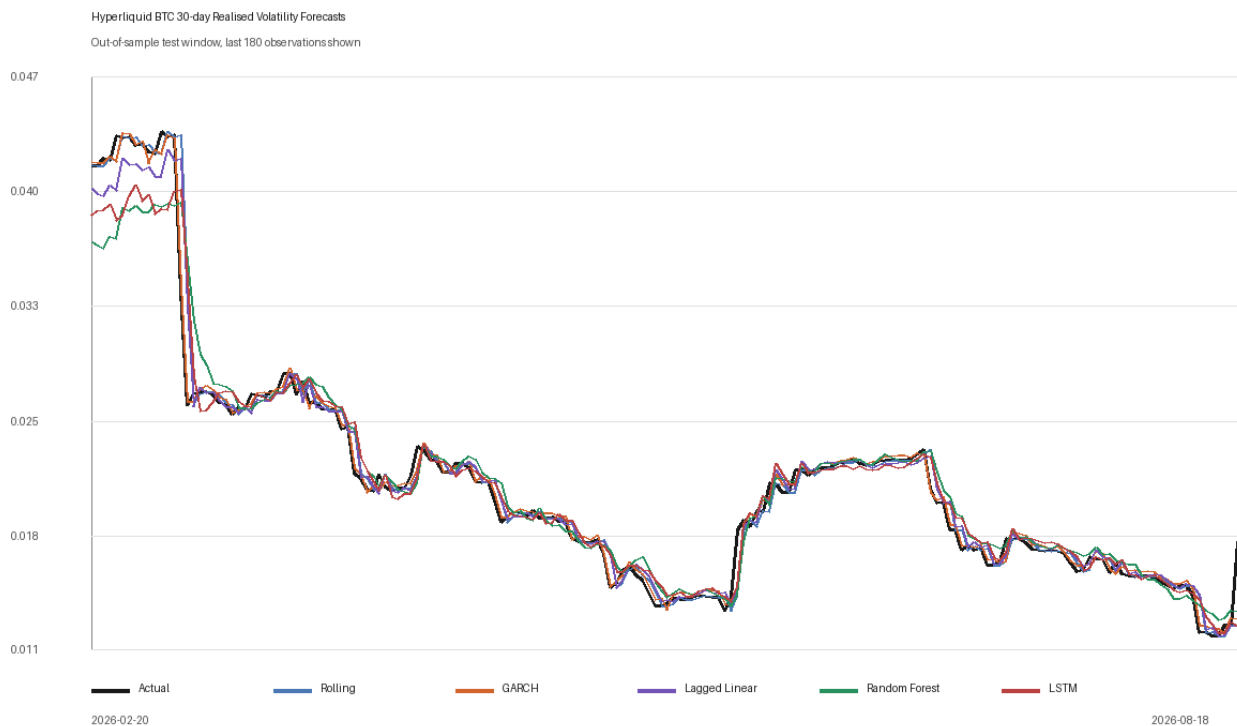


5. Risk Assessment

Risk	Likelihood	Impact	Mitigation
Scope becomes too broad	Medium	High	Focus on one BTC perpetual-futures market and one next-day proxy update
Open or malformed candles enter modelling	Low after controls	High	Require completed timestamps, strict cadence, valid OHLC and positive prices
Time-series leakage or date misalignment	Low after controls	High	Use a fixed chronological cutoff, date-keyed forecasts and regression tests
GARCH target conversion is implemented incorrectly	Low after controls	High	Derive the conversion, test the recursion and retain an analytic sensitivity
RF/LSTM tuning uses the final test	Low after controls	High	Select only on the chronological tail of training data and export selection tables
Pipelines are mistaken for an architecture-only experiment	Medium	High	Disclose features, sample sizes and transformations; bound the conclusion
The proxy is treated as true latent volatility	Medium	Medium	State the 29-return overlap, add QLIKE and retain proxy limitations
One history is treated as universal evidence	Medium	Medium	Use windows, halves, folds, regimes, refresh and block bootstrap; limit claims
AI-supported work is disclosed inaccurately	Medium	High	Require a candidate-specific disclosure and keep unauthenticated fields blank

The maintained full register is in `appendix/risk-assessment.md`.

6. Data Charts



Rank	Model	RMSE	QLIKE
1	GARCH(1,1)	0.00098301	0.00445845
2	Lagged linear regression	0.00137492	0.00724584
3	Rolling historical volatility	0.00139570	0.00754794
4	LSTM	0.00168735	0.00897661
5	Random Forest	0.00211129	0.01182354

The chart must be read in light of the overlapping target. On the final 19 August shock target, Random Forest had the smallest one-day error, but all five models underpredicted the jump and Random Forest remained fifth overall. Full tables and robustness outputs are indexed in `appendix/model-results-summary.md` and stored under `code/outputs/`.